

Mayur Naik

Misra Family Professor
Department of Computer and Information Science
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1. RESEARCH INTERESTS

My research spans the areas of programming languages and artificial intelligence. The overarching goal of my research is to improve programmer productivity and software quality.

2. EDUCATION

Degree	Year	University	Field
Ph.D.	2008	Stanford University Stanford, CA <i>Dissertation:</i> Effective Static Race Detection for Java <i>Advisor:</i> Alexander Aiken	Computer Science
M.S.	2003	Purdue University W. Lafayette, IN <i>Dissertation:</i> A Type System Equivalent to Model Checking <i>Advisor:</i> Jens Palsberg	Computer Science
B.E.	1999	Birla Institute of Technology and Science Pilani, India	Computer Science

3. APPOINTMENTS

Title	Organization	Years
Misra Family Professor	University of Pennsylvania	07-2024 – present
Professor	University of Pennsylvania	07/2020 – present
Associate Professor	University of Pennsylvania	08/2016 – 06/2020
Assistant Professor	Georgia Institute of Technology	07/2011 – 08/2016
Research Scientist	Intel Research	10/2007 – 05/2011

4. HONORS AND AWARDS

4.1. National Awards

- NSF CAREER Award, 2013.
- Microsoft Software Engineering Innovation Foundation (SEIF) Award, 2012.
- Microsoft Research Graduate Fellowship, 2004-2005.

4.2. Publication Awards

- Test-of-Time Paper Award, 2023.
In 21st ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE'13) for the paper titled “Dynodroid: An Input Generation System for Android Apps”.
- Test-of-Time Paper Award, 2022.
In 20th ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE'12) for the paper titled “Automated Concolic Testing of Smartphone Apps”.
- Test-of-Time Paper Award, 2021.
In 6th European Conference on Computer Systems (EuroSys'11) for the paper titled “CloneCloud: Elastic Execution between Mobile Device and Cloud”.
- Distinguished Paper Award, 2019.
In 40th ACM Conference on Programming Language Design and Implementation (PLDI'19) for the paper titled “Continuously Reasoning about Programs using Differential Bayesian Inference”.
- Distinguished Paper Award, 2015.
In 23rd ACM Symposium on Foundations of Software Engineering (FSE'15) for the paper titled “A User-Guided Approach to Program Analysis”.
- Distinguished Paper Award, 2014.
In 35th ACM Conference on Programming Language Design and Implementation (PLDI'14) for the paper titled “On Abstraction Refinement for Program Analyses in Datalog”.
- Distinguished Artifact Award, 2013.
For the paper titled “Dynodroid: An Input Generation System for Android Apps” in Proceedings of the 21st ACM Symposium on Foundations of Software Engineering (FSE'13).
- Distinguished Paper Award, 2009.
For the paper titled “Effective Static Deadlock Detection” in Proceedings of the 31st International Conference on Software Engineering (ICSE'09).

4.3. University Awards

- Outstanding Junior Faculty Research Award, Georgia Tech, 2016.
Awarded for “the quality of publications and the significance and impact of research.”
- Lockheed-Martin Teaching Excellence Award, Georgia Tech, 2015.
Awarded for “extraordinary effectiveness in classroom teaching, educational innovations, inspiration transmitted to students, direct impact and involvement with students, and impact on the postgraduate success of students.”

5. PUBLICATIONS

5.1. Books

1. Ziyang Li, Jiani Huang, Jason Liu, Mayur Naik. Neurosymbolic Programming in Scallop: Principles and Practice. In *Foundations and Trends in Programming Languages*. NOW Publishers, 2024.

5.2. Refereed Conference Papers

Note: Author names annotated * indicate co-first authorship.

1. Claire Wang, Ziyang Li, Saikat Dutta, Mayur Naik. QLCoder: A Query Synthesizer For Static Analysis of Security Vulnerabilities. In *International Conference on Learning Representations (ICLR'26)*, April 2026.

2. Paul Biberstein, Ziyang Li, Joseph Devietti, Mayur Naik. Lobster: A GPU-Accelerated Framework for Neurosymbolic Programming. In *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'26)*, March 2026.
3. Jiani Huang, Amish Sethi*, Matthew Kuo*, Mayank Keoliya, Neelay Velingker, JungHo Jung, Ser-Nam Lim, Ziyang Li, Mayur Naik. ESCA: Contextualizing Embodied Agents via Scene-Graph Generation. In *Conference on Neural Information Processing Systems (NeurIPS'25)*, December 2025. **Spotlight Paper**.
4. Adam Stein*, Neelay Velingker*, Mayur Naik, Eric Wong. Once Upon an Input: Reasoning via Per-Instance Program Synthesis. In *Conference on Neural Information Processing Systems (NeurIPS'25)*, December 2025.
5. Aaditya Naik, Jason Liu, Claire Wang, Amish Sethi, Saikat Dutta, Mayur Naik, Eric Wong. Dolphin: A Programmable Framework for Scalable Neurosymbolic Learning. In *International Conference on Machine Learning (ICML'25)*, July 2025.
6. Jiani Huang, Ziyang Li, Mayur Naik, Ser Nam Lim. LASER: A Neuro-Symbolic Framework for Learning Spatial-Temporal Scene Graphs with Weak Supervision. In *International Conference on Learning Representations (ICLR'25)*, April 2025.
7. Ziyang Li, Saikat Dutta, Mayur Naik. IRIS: LLM-Assisted Static Analysis for Detecting Security Vulnerabilities. In *International Conference on Learning Representations (ICLR'25)*, April 2025.
8. Avishree Khare*, Saikat Dutta*, Ziyang Li, Alaia Solko-Breslin, Rajeev Alur, Mayur Naik. Understanding the Effectiveness of Large Language Models in Detecting Security Vulnerabilities. In *International Conference on Software Testing, Verification and Validation (ICST'25)*, March 2025.
9. Alaia Solko-Breslin, Seewon Choi, Ziyang Li, Neelay Velingker, Rajeev Alur, Mayur Naik, Eric Wong. Data-Efficient Learning with Neural Programs. In *Conference on Neural Information Processing Systems (NeurIPS'24)*, December 2024.
10. Ziyang Li*, Chang Zhu*, Anton Xue, Ati Priya Bajaj, William Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, Adam Doupé, Mayur Naik, Yan Shoshitaishvili, Ruoyu Wang, Aravind Machiry. TYGR: Type Inference on Stripped Binaries using Graph Neural Networks. In *USENIX Security Symposium (Security'24)*, August 2024.
11. Adam Stein, Aaditya Naik, Yinjun Wu, Mayur Naik, Eric Wong. Towards Compositionality in Concept Learning. In *International Conference on Machine Learning (ICML'24)*, July 2024.
12. Yinjun Wu, Mayank Keoliya, Kan Chen, Neelay Velingker, Ziyang Li, Emily J Getzen, Qi Long, Mayur Naik, Ravi B Parikh, Eric Wong. DISCRET: Synthesizing Faithful Explanations For Treatment Effect Estimation. In *International Conference on Machine Learning (ICML'24)*, July 2024.
13. Aaditya Naik, Adam Stein, Yinjun Wu, Mayur Naik, Eric Wong. TorchQL: A Programming Framework for Integrity Constraints in Machine Learning. In *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'24)*, October 2024.
14. Ziyang Li, Jiani Huang, Jason Liu, Felix Zhu, Eric Zhao, William Dodds, Neelay Velingker, Rajeev Alur, Mayur Naik. Relational Programming with Foundation Models. In *AAAI Conference on Artificial Intelligence (AAAI'24)*, February 2024.
15. Aaditya Naik, Aalok Thakkar, Adam Stein, Rajeev Alur, Mayur Naik. Relational Query Synthesis $\bowtie$ Decision Tree Learning. In *International Conference on Very Large Data Bases (VLDB'24)*, August 2024.
16. Aalok Thakkar, Nathaniel Sands, George Petrou, Rajeev Alur, Mayur Naik, Mukund Raghothaman. Mobius: Synthesizing Relational Queries with Recursive and Invented Predicates. In *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'23)*, October 2023.
17. Hanlin Zhang*, Jiani Huang*, Ziyang Li, Mayur Naik, Eric Xing. Improved Logical Reasoning of Language Models via Differentiable Symbolic Programming. In *Findings of Association for Computational Linguistics (ACL'23)*, July 2023.

18. Aaditya Naik, Yinjun Wu, Mayur Naik, Eric Wong. Do Machine Learning Models Learn Statistical Rules Inferred from Data? In *International Conference on Machine Learning (ICML'23)*, July 2023.
19. Ziyang Li, Jiani Huang, Mayur Naik. Scallop: A Language for Neurosymbolic Programming. In *ACM Conference on Programming Language Design and Implementation (PLDI'23)*, June 2023.
20. Yinjun Wu, Adam Stein, Jacob Gardner, Mayur Naik. Learning to Select Pivotal Samples for Meta Re-weighting. In *AAAI Conference on Artificial Intelligence (AAAI'23)*, February 2023. **Oral Presentation.**
21. Elizabeth Dinella, Todd Mytkowicz, Alexey Svyatkovskiy, Christian Bird, Mayur Naik, Shuvendu Lahiri. Deep-Merge: Learning to Merge Programs. In *ACM Symposium on Foundations of Software Engineering (FSE'22)*, November 2022.
22. Pardis Pashakhanloo, Aravind Machiry, Hyonyoung Choi, Anthony Canino, Kihong Heo, Insup Lee, Mayur Naik. PacJam: Securing Dependencies Continuously via Package-Oriented Debloating. In *ACM ASIA Conference on Computer and Communications Security (ASIACCS'22)*, May 2022.
23. Pardis Pashakhanloo, Aaditya Naik, Yuepeng Wang, Hanjun Dai, Petros Maniatis, Mayur Naik. CodeTrek: Flexible Modeling of Code using an Extensible Relational Representation. In *International Conference on Learning Representations (ICLR'22)*, April 2022.
24. Jiani Huang*, Ziyang Li*, Binghong Chen, Karan Samel, Mayur Naik, Le Song, Xujie Si. Scallop: From Probabilistic Deductive Databases to Scalable Differentiable Reasoning. In *Conference on Neural Information Processing Systems (NeurIPS'21)*, December 2021.
25. Aaditya Naik, Jonathan Mendelson, Nate Sands, Yuepeng Wang, Mayur Naik, Mukund Raghothaman. Sporq: An Interactive Environment for Exploring Code Using Query-by-Example. In *ACM Symposium on User Interface Software and Technology (UIST'21)*, October 2021.
26. Aalok Thakkar, Aaditya Naik, Nathaniel Sands, Rajeev Alur, Mayur Naik, Mukund Raghothaman. Example-Guided Synthesis of Relational Queries. In *ACM Conference on Programming Language Design and Implementation (PLDI'21)*, June 2021.
27. Ziyang Li, Aravind Machiry, Binghong Chen, Ke Wang, Mayur Naik, Le Song. Arbitrar: User-Guided API Misuse Detection. In *IEEE Symposium on Security and Privacy (S&P'21)*, May 2021.
28. Jonathan Mendelson*, Aaditya Naik*, Mukund Raghothaman, Mayur Naik. GenSynth: Synthesizing Datalog Programs without Language Bias. In *AAAI Conference on Artificial Intelligence (AAAI'21)*, February 2021.
29. Jiani Huang, Calvin Smith, Osbert Bastani, Rishabh Singh, Aws Albarghouthi, Mayur Naik. Generating Programmatic Referring Expressions via Program Synthesis. In *International Conference on Machine Learning (ICML'20)*, July 2020.
30. Xujie Si*, Aaditya Naik*, Hanjun Dai, Mayur Naik, Le Song. Code2Inv: A Deep Learning Framework for Program Verification. In *Conference on Computer Aided Verification (CAV'20)*, June 2020.
31. Elizabeth Dinella*, Hanjun Dai*, Ziyang Li, Mayur Naik, Le Song, Ke Wang. Hoppity: Learning Graph Transformations to Detect and Fix Bugs in Programs. In *International Conference on Learning Representations (ICLR'20)*, April 2020. **Spotlight Paper.**
32. Mukund Raghothaman, Jonathan Mendelson, David Zhao, Mayur Naik, Bernhard Scholz. Provenance-Guided Synthesis of Datalog Programs. In *ACM Symposium on Principles of Programming Languages (POPL'20)*, January 2020.
33. Xujie Si*, Mukund Raghothaman*, Kihong Heo, Mayur Naik. Synthesizing Datalog Programs using Numerical Relaxation. In *International Joint Conferences on Artificial Intelligence (IJCAI'19)*, August 2019.
34. Kihong Heo*, Mukund Raghothaman*, Xujie Si, Mayur Naik. Continuously Reasoning about Programs using Differential Bayesian Inference. In *ACM Conference on Programming Language Design and Implementation (PLDI'19)*, June 2019. **Distinguished Paper Award.**
35. Halley Young, Osbert Bastani, Mayur Naik. Learning Neurosymbolic Generative Models via Program Synthe-

- sis. In *International Conference on Machine Learning (ICML'19)*, June 2019.
36. Xujie Si*, Yuan Yang*, Hanjun Dai, Mayur Naik, Le Song. Learning a Meta-Solver for Syntax-Guided Program Synthesis. In *International Conference on Learning Representations (ICLR'19)*, May 2019.
 37. Xujie Si*, Hanjun Dai*, Mukund Raghothaman, Mayur Naik, Le Song. Learning Loop Invariants for Program Verification. In *Conference on Neural Information Processing Systems (NeurIPS'18)*, December 2018. **Spotlight Paper.**
 38. Xujie Si*, Woosuk Lee*, Richard Zhang, Aws Albarghouthi, Paris Koutris, Mayur Naik. Syntax-Guided Synthesis of Datalog Programs. In *ACM Symposium on Foundations of Software Engineering (FSE'18)*, November 2018.
 39. Kihong Heo*, Woosuk Lee*, Pardis Pashakhanloo, Mayur Naik. Effective Program Debloating via Reinforcement Learning. In *ACM Conference on Computer and Communications Security (CCS'18)*, October 2018.
 40. Mukund Raghothaman*, Sulekha Kulkarni*, Kihong Heo, Mayur Naik. User-Guided Program Reasoning Using Bayesian Inference. In *ACM Conference on Programming Language Design and Implementation (PLDI'18)*, June 2018.
 41. Woosuk Lee, Kihong Heo, Rajeev Alur, Mayur Naik. Accelerating Search-Based Program Synthesis Using Learned Probabilistic Models. In *ACM Conference on Programming Language Design and Implementation (PLDI'18)*, June 2018.
 42. Xin Zhang, Radu Grigore, Xujie Si, Mayur Naik. Effective Interactive Resolution of Static Analysis Alarms. In *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'17)*, October 2017.
 43. Aws Albarghouthi, Paraschos Koutris, Mayur Naik, Calvin Smith. Constraint-Based Synthesis of Datalog Programs. In *International Conference on Principles and Practice of Constraint Programming (CP'17)*, September 2017.
 44. Sulekha Kulkarni, Ravi Mangal, Xin Zhang, Mayur Naik. Accelerating Program Analyses by Cross-Program Training. In *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'16)*, November 2016.
 45. Xujie Si, Xin Zhang, Vasco Manquinho, Mikolas Janota, Alexey Ignatiev, Mayur Naik. On Incremental Core-Guided MaxSAT Solving. In *International Conference on Principles and Practice of Constraint Programming (CP'16)*, September 2016.
 46. Insu Yun, Changwoo Min, Xujie Si, Yeongjin Jang, Taesoo Kim, Mayur Naik. APISan: Sanitizing API Usages through Semantic Cross-checking. In *USENIX Security Symposium (Security'16)*, August 2016.
 47. Ravi Mangal, Xin Zhang, Aditya Kamath, Aditya V. Nori, Mayur Naik. Scaling Relational Inference Using Proofs and Refutations. In *AAAI Conference on Artificial Intelligence (AAAI'16)*, February 2016.
 48. Xin Zhang, Ravi Mangal, Aditya V. Nori, Mayur Naik. Query-Guided Maximum Satisfiability. In *ACM Symposium on Principles of Programming Languages (POPL'16)*, January 2016.
 49. Ravi Mangal, Xin Zhang, Aditya V. Nori, Mayur Naik. Volt: A Lazy Grounding Framework for Solving Very Large MaxSAT Instances. In *International Conference on Theory and Applications of Satisfiability Testing (SAT'15)*, September 2015.
 50. Ravi Mangal, Xin Zhang, Aditya V. Nori, Mayur Naik. A User-Guided Approach to Program Analysis. In *ACM Symposium on Foundations of Software Engineering (FSE'15)*, August 2015. **Distinguished Paper Award.**
 51. Jongse Park, Hadi Esmaeilzadeh, Xin Zhang, Mayur Naik, William Harris. FlexJava: Language Support for Safe and Modular Approximate Programming. In *ACM Symposium on Foundations of Software Engineering (FSE'15)*, August 2015.
 52. Ghila Castelnovo, Mayur Naik, Noam Rinetzky, Mooly Sagiv, Hongseok Yang. Modularity in lattices: A case study on the correspondence between top-down and bottom-up analysis. In *International Static Analysis*

Symposium (SAS'15), September 2015.

53. Cong Shi, Karim Habak, Pranesh Pandurangan, Mostafa Ammar, Mayur Naik, Ellen Zegura. COSMOS: Computation Offloading as a Service for Mobile Devices. In *ACM Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc'14)*, August 2014.
54. Xin Zhang, Ravi Mangal, Mayur Naik, Radu Grigore, Hongseok Yang. On Abstraction Refinement for Program Analyses in Datalog. In *ACM Conference on Programming Language Design and Implementation (PLDI'14)*, June 2014. **Distinguished Paper Award.**
55. Xin Zhang, Ravi Mangal, Mayur Naik, Hongseok Yang. Hybrid Top-Down and Bottom-Up Interprocedural Analysis. In *ACM Conference on Programming Language Design and Implementation (PLDI'14)*, June 2014.
56. Ravi Mangal, Mayur Naik, Hongseok Yang. A Correspondence between Two Approaches to Interprocedural Analysis in the Presence of Join. In *European Symposium on Programming (ESOP'14)*, April 2014.
57. Aravind Machiry, Rohan Tahlilani, Mayur Naik. Dynodroid: An Input Generation System for Android Apps. In *ACM Symposium on Foundations of Software Engineering (FSE'13)*, August 2013. **Distinguished Artifact Award.**
58. Yongin Kwon, Sangmin Lee, Hayoon Yi, Donghyun Kwon, Seungjun Yang, Byung-Gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek. Mantis: Automatic Performance Prediction for Smartphone Applications. In *USENIX Annual Technical Conference (ATC'13)*, June 2013.
59. Xin Zhang, Mayur Naik, Hongseok Yang. Finding Optimum Abstractions in Parametric Dataflow Analysis. In *ACM Conference on Programming Language Design and Implementation (PLDI'13)*, June 2013.
60. Saswat Anand, Mayur Naik, Hongseok Yang, Mary Jean Harrold. Automated Concolic Testing of Smartphone Apps. In *ACM Symposium on Foundations of Software Engineering (FSE'12)*, November 2012.
61. Mayur Naik, Hongseok Yang, Ghila Castelnovo, Mooly Sagiv. Abstractions from Tests. In *ACM Symposium on Principles of Programming Languages (POPL'12)*, January 2012.
62. Percy Liang and Mayur Naik. Scaling Abstraction Refinement via Pruning. In *ACM Conference on Programming Language Design and Implementation (PLDI'11)*, June 2011.
63. Byung-Gon Chun, Sunghwan Ihm, Petros Maniatis, Mayur Naik, Ashwin Patti. CloneCloud: Elastic Execution between Mobile Device and Cloud. In *European Conference on Computer Systems (EuroSys'11)*, April 2011.
64. Percy Liang, Omer Tripp, Mayur Naik. Learning Minimal Abstractions. In *ACM Symposium on Principles of Programming Languages (POPL'11)*, January 2011.
65. Ling Huang, Jinzhu Jia, Bin Yu, Byung-Gon Chun, Petros Maniatis, Mayur Naik. Predicting Execution Time of Computer Programs Using Sparse Polynomial Regression. In *Conference on Neural Information Processing Systems (NIPS'10)*, December 2010.
66. Pallavi Joshi, Mayur Naik, Koushik Sen, David Gay. An Effective Dynamic Analysis for Detecting Generalized Deadlocks. In *ACM Symposium on Foundations of Software Engineering (FSE'10)*, November 2010.
67. Percy Liang, Omer Tripp, Mayur Naik, Mooly Sagiv. A Dynamic Evaluation of the Precision of Static Heap Abstractions. In *ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA'10)*, October 2010.
68. Pallavi Joshi, Mayur Naik, Chang-Seo Park, Koushik Sen. CalFuzzer: An Extensible Active Testing Framework for Concurrent Programs. In *Conference on Computer Aided Verification (CAV'09)*, June 2009.
69. Zachary Anderson, David Gay, Mayur Naik. Lightweight Annotations for Controlling Sharing in Concurrent Data Structures. In *ACM Conference on Programming Language Design and Implementation (PLDI'09)*, June 2009.
70. Pallavi Joshi, Chang-Seo Park, Koushik Sen, Mayur Naik. A Randomized Dynamic Program Analysis Technique for Detecting Real Deadlocks. In *ACM Conference on Programming Language Design and Implementation (PLDI'09)*, June 2009.

71. Mayur Naik, Chang-Seo Park, Koushik Sen, David Gay. Effective Static Deadlock Detection. In *International Conference on Software Engineering (ICSE'09)*, May 2009.
72. Mayur Naik and Alex Aiken. Conditional Must Not Aliasing for Static Race Detection. In *ACM Symposium on Principles of Programming Languages (POPL'07)*, January 2007.
73. Mayur Naik, Alex Aiken, John Whaley. Effective Static Race Detection for Java. In *ACM Conference on Programming Language Design and Implementation (PLDI'06)*, June 2006.
74. Alice X. Zheng, Michael I. Jordan, Ben Liblit, Mayur Naik, Alex Aiken. Statistical Debugging: Simultaneous Isolation of Multiple Bugs. In *International Conference on Machine Learning (ICML'06)*, June 2006.
75. Ben Liblit, Mayur Naik, Alice X. Zheng, Alex Aiken, Michael I. Jordan. Scalable Statistical Bug Isolation. In *ACM Conference on Programming Language Design and Implementation (PLDI'05)*, June 2005.
76. Mayur Naik and Jens Palsberg. A Type System Equivalent to a Model Checker. In *European Symposium on Programming (ESOP'05)*, April 2005.
77. Thomas Ball, Mayur Naik, Sriram K. Rajamani. From Symptom to Cause: Localizing Errors in Counterexample Traces. In *ACM Symposium on Principles of Programming Languages (POPL'03)*, January 2003.
78. Mayur Naik and Jens Palsberg. Compiling with Code-Size Constraints. In *Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES'02)*, June 2002.

5.3. Journal Articles

1. Elizabeth Dinella, Todd Mytkowicz, Alexey Svyatkovskiy, Christian Bird, Mayur Naik, Shuvendu K. Lahiri. DeepMerge: Learning to Merge Programs. *IEEE Transactions on Software Engineering*, 2022.
2. Haoxian Chen, Chenyuan Wu, Andrew Zhao, Mukund Raghothaman, Mayur Naik, Boon Thau Loo. Synthesizing Formal Network Specifications from Input-Output Examples. *IEEE/ACM Transactions on Networking*, 2022.
3. Yongin Kwon, Byung-Gon Chun, Hayoon Yi, Donghyun Kwon, Seungjun Yang, Sangmin Lee, Ling Huang, Petros Maniatis, Mayur Naik, Yunheung Paek. Mantis: Efficient Predictions of Execution Time, Energy Usage, Memory Usage and Network Usage on Smart Mobile Devices. *IEEE Transactions on Mobile Computing*, 14(10):2059–2072, 2015.
4. David Gay, Joel Galenson, Mayur Naik, Kathy Yelick. Yada: Straightforward Parallel Programming. *Parallel Computing*, 37(9):592–609, 2011.
5. Mayur Naik and Jens Palsberg. A Type System Equivalent to a Model Checker. *ACM Transactions on Programming Languages and Systems*, 30(5), 2008.
6. Mayur Naik and Jens Palsberg. Compiling with Code-Size Constraints. *ACM Transactions on Embedded Computing Systems*, 3(1):163–181, 2004.

5.4. Invited Papers

1. Mayur Naik. Rethinking Static Analysis by Combining Discrete and Continuous Reasoning. In *International Static Analysis Symposium (SAS'19)*, October 2019.
2. Xujie Si, Xin Zhang, Radu Grigore, Mayur Naik. Maximum Satisfiability in Program Analysis: Applications and Techniques. In *International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI'18)*, January 2018.
3. Xujie Si, Xin Zhang, Radu Grigore, Mayur Naik. Maximum Satisfiability in Software Analysis: Applications and Techniques. In *International Conference on Computer Aided Verification (CAV'17)*, June 2017.

5.5. Book Chapters

1. Jens Palsberg and Mayur Naik. ILP-based Resource-aware Compilation. In *Multiprocessor Systems-on-Chips*. Morgan Kaufmann, 2004.

5.6. Edited Volumes

1. Jennifer Sartor, Mayur Naik, and Chris Rossbach. Proceedings of the 15th ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments (VEE 2019), Providence, RI, USA. Co-located with ASPLOS'19, April 2019.
2. Anders Moeller and Mayur Naik. Proceedings of the 4th ACM SIGPLAN International Workshop on the State Of The Art in Java Program Analysis (SOAP 2015), Portland, Oregon, USA. Co-located with PLDI'15, June 2015.

6. GRANTS AND CONTRACTS

6.1. As Principal Investigator

1. Mayur Naik. "Neuro-Symbolic Analysis of Source Code to Identify and Prioritize Security Vulnerability Alarms". AFRL #FA8750-26-2-B004, \$1,780,468, April 2026 (3 years).
2. Mayur Naik, Eric Wong, Rajeev Alur. "SHF: Medium: Scallop: A Neurosymbolic Programming Framework for Combining Logic with Deep Learning". NSF CCF #2313010, \$1,200,000, Oct 2023 (4 years).
3. Mayur Naik, Rajeev Alur, Mukund Raghothaman. "SHF: Medium: Synthesis of Logic Programs for Democratizing Program Analysis". NSF CCF #2107429, \$1,080,000, May 2021 (4 years).
4. Mayur Naik. "Drake: Signature-Guided Detection of Bugs and Vulnerabilities". Air Force Research Laboratory #FA8750-20-2-0501, \$170,000, October 2019 (1 year).
5. Mayur Naik. "CODA: A Deep Reinforcement Learning Framework for Building Code Assistants". Facebook Research Award, \$50,000, May 2019 (1 year).
6. Mayur Naik. "An AI-based System for Continuously Discovering Software Bugs". Facebook Research Award, \$50,000, November 2018 (1 year).
7. Mayur Naik and Le Song. "Synergies between Program Synthesis and Neural Learning of Graph Structures". NSF CCF #1836936, \$900,000, September 2018 (4 years).
8. Mayur Naik, Rajeev Alur, Insup Lee, Oleg Sokolsky, Boon Thau Loo. "ASPIRE: Automatically Subsetting Protocol Implementations Reliably and Efficiently". ONR #N00014-18-1-2021, \$6,148,632, January 2018 (5 years).
9. Mayur Naik. "SHF: Small: New Frontiers in Constraint-Based Program Analysis". NSF CCF #1526270, \$450,000, September 2015 (3 years).
10. Mayur Naik, Molham Aref, Shan Shan Huang, Jens Palsberg, Tielei Wang. "PetaBlox: Large-Scale Software Analysis and Analytics Using Datalog". DARPA #FA8750-15-2-0009, \$3,640,660, October 2014 (5 years).
11. Mayur Naik. "CAREER: Adaptive Large-Scale Program Analysis". NSF CCF #1253867, \$484,402, January 2013 (5 years).
12. Mayur Naik. "VIVAS: Verification and Validation of Autonomous Systems". MIT Lincoln Lab, \$50,000, November 2016 (1 year).
13. Mayur Naik. "Automated Scalable Testing of Mobile Apps Using Z3". Microsoft Software Engineering Innovation Foundation Award, \$25,000, March 2012 (1 year).
14. Mayur Naik. "Enhancing Mobile App Quality via Program Analysis". Google Faculty Research Award, \$52,786, December 2011 (1 year).

6.2. As Co-Principal Investigator

1. Qi Long (PI), Mayur Naik, Ravi Parikh, Marilyn Schapira. “Advancing Neurosymbolic AI for Smart and Trustworthy Healthcare”. NIH #1-R01-EB-036016, \$2,564,852, August 2025 (4 years).
2. Rajeev Alur (PI), Rajat Deo, Sameed Khatana, Qi Long, Mayur Naik, Ravi Parikh, Payal Shah, Gary Weissman, Eric Wong. “Safe and Explainable AI enabled Decision Making for Personalized Clinical Decision Support”. ARPA-H #D24AC00253-00, \$6,850,181, September 2024 (4 years).
3. Shankar Sastry (PI), Sergey Levine, Sanjit Seshia, Claire Tomlin, Somil Bansal, Mayur Naik. “Provably Correct Design of Adaptive Hybrid Neuro-Symbolic Cyber Physical Systems”. DARPA #FA8750-23-C-0080, \$5,398,530, June 2023 (4 years).
4. Vincent Liu (PI), Rajeev Alur, Sebastian Angel, Mayur Naik. “FMitF: Track I: Automatic Migration to Serverless Infrastructure, Correctly and Efficiently”. NSF CCF # 2124184, \$750,000, July 2021 (3 years).
5. Dan Roth (PI), Osbert Bastani, Chris Callison-Burch, Mayur Naik, Irfan Essa, Le Song, Kira Zsolt. “FLASH: Fast Learning via Auxiliary signals, Structured knowledge, and Human expertise”. DARPA #FA8750-19-2-0201, \$3,349,000, September 2019 (3 years).
6. Taesoo Kim (PI), Emmanouil Antonakakis, Mayur Naik, Wenke Lee. “SaTC-EDU: EAGER: Big Data and Security: Educating The Next-Generation Security Analysts”. NSF DGE #1500084, \$300,000, June 2015 (3 years).
7. Mostafa Ammar (PI), Irfan Issa, Mayur Naik, Ellen Zegura. “NeTS: Medium: Mobile Computing over Intermittently Connected Networks”. NSF CNS #1161879, \$695,000, August 2012 (3 years).
8. Hyesoon Kim (PI), Mayur Naik, Santosh Pande. “Smart Offloading Algorithms and Profiling Techniques for Cloud Computing”. Samsung, \$185,000, May 2012 (1 year).
9. Alex Aiken (PI), Isil Dillig, Thomas Dillig, John Mitchell, Mayur Naik. “STAMP: Static Analysis of Mobile Programs”. DARPA #FA8750-12-2-0020, \$3,929,766, January 2012 (4 years).

7. STUDENTS

7.1. Ph.D. Students

Current

1. **Paul Biberstein**, 2023–present
2. **Jiani Huang**, 2018–present
3. **Mayank Keoliya**, 2023–present
4. **Aaditya Naik**, 2020–present
5. **Adam Stein**, 2021–present
6. **Neelay Velingker**, 2024–present
7. **Claire Wang**, 2025–present
8. **Zhiqiu (Oscar) Xu**, 2024–present

Graduated

1. **Xin Zhang**, 2011–2017
Thesis Topic: Combining Logical and Probabilistic Reasoning in Program Analysis
Facebook Fellowship (2015-16) and Georgia Tech College of Computing’s 2017 Outstanding Graduate Research Award

Current Position: Assistant Professor, Department of Computer Science and Technology, Peking University, China

2. **Xujie Si**, 2014–2020
Thesis Topic: Learning-Aided Program Synthesis and Verification
Canada CIFAR AI Chair
Current Position: Assistant Professor, Department of Computer Science, University of Toronto
3. **Sulekha Kulkarni**, 2014–2020
Thesis Topic: Effective Program Reasoning using Bayesian Inference
Current Position: Principal Software Engineer, Microsoft
4. **Pardis Pashakhanloo**, 2018–2023
Thesis Topic: Integrating Declarative Static Analysis with Neural Models of Code
Current Position: Applied Scientist, Amazon
5. **Aalok Thakkar**, 2018–2023
Co-advised with Rajeev Alur
Thesis Topic: Example-Guided Synthesis of Relational Queries
Current Position: Assistant Professor, Ashoka University
6. **Elizabeth Dinella**, 2018–2023
Thesis Topic: Neural Inference of Program Specifications (Winner of 2025 SIGSOFT Outstanding Dissertation Award)
2025 ACM SIGSOFT Outstanding Dissertation Award
Current Position: Assistant Professor, Bryn Mawr College
7. **Ziyang Li**, 2019–2025
Thesis Topic: Neurosymbolic Programming in Scallop: Design, Implementation, and Applications
2026 Morris and Dorothy Rubinoff Dissertation Award
Current Position: Assistant Professor, Johns Hopkins University

7.2. Postdoctoral Scholars

1. **Woosuk Lee**, 2016–2018
Current Position: Assistant Professor, College of Computing, Hanyang University, Korea
2. **Kihong Heo**, 2017–2019
Current Position: Assistant Professor, School of Computing, KAIST, Korea
3. **Mukund Raghothaman**, 2017–2019
Current Position: Assistant Professor, Department of Computer Science, University of Southern California
4. **Aravind Machiry**, 2020–2020
Current Position: Assistant Professor, Department of Electrical and Computer Engineering, Purdue University
5. **Yuepeng Wang**, 2020–2021
Co-advised with Rajeev Alur
Current Position: Assistant Professor, School of Computing Science, Simon Fraser University, Canada
6. **Yinjun Wu**, 2022–2024
Co-advised with Qi Long
Current Position: Assistant Professor, Department of Computer Science and Technology, Peking University, China
7. **Saikat Dutta**, 2023–2024
Current Position: Assistant Professor, Cornell University